



Zircon

△ **Cushion-cut**, 10-carat zircon from Myanmar

Some zircon material is 4.4 billion years old, making it the oldest-known mineral on Earth. It is a colourful gem with high refraction and fire. Colourless zircon is known for its luminescence and reflective flashes of multicoloured light, and is often used in jewellery as a substitute for diamonds. Vibrant blue zircons are produced by heat-treating the more common brown stones. The mineral sometimes contains traces of uranium and thorium, and this natural radioactivity can disrupt the crystal structure, causing changes to colour, density, RI, and double refraction.

Specification

Chemical name Zirconium silicate | **Formula** ZrSiO_4
Colours Reddish brown, yellow, green, blue, grey, colourless
Structure Tetragonal | **Hardness** 6.5–7.5 | **SG** 3.9–4.7
RI 1.81–2.02 | **Lustre** Vitreous to brilliant sheen | **Streak** White | **Locations** Australia, Myanmar, Cambodia, Tanzania

Rough



Zircon crystals | Zircon crystals can form in many different types of rock. This specimen is set on a groundmass of pegmatite, a common zircon occurrence.



Classic crystal | This reddish-brown specimen is a typical example of a tetragonal prism capped on both ends by fine pyramid-shaped terminations.



Raw colours | This group of water-rounded, yellow to red-brown zircons demonstrates the variance between specimens when found in their natural state.

Colours and cuts



Blue pendeloque | At 15 carats, this magnificent zircon has been heat-treated to intensify its colour. It is pendeloque-cut and displays superb dispersion.